Topic 00: Infosys Puzzle Solving Master Formulas, Axioms & Shortcuts Cheatsheet

> **Target Exam:** Infosys System Engineer (SE), Digital Specialist Engineer (DSE), Specialist Programmer (SP)
> **Domain:** Mathematical Puzzles, Matrix Grids, Geometric Figures, Alphabet Positional Reasoning
> **Status:** Production-Ready Engineering Cheatsheet

1. Executive Summary & Core Axioms

In the Infosys recruitment examination, the **Puzzle Solving** section assesses a candidate's high-speed algebraic pattern recognition, discrete logic deduction, and spatial arithmetic acuity. Puzzles are presented in two distinct formats:

1. **Matrix Number Grids:** 3×3, 4×3, or 4×4 numeric matrices with one or more unknown values (`?`, `X`, `Y`).
2. **Geometric & Polygon Figures:** Triangles, circles, propellers, ovals, and multi-segmented boxes with peripheral inputs mapping to an internal focal value.

The Fundamental Axiom of Pattern Invariance

Every valid puzzle in the Infosys examination satisfies a **single, deterministic algebraic rule** across all reference components (rows, columns, or figures). If a candidate rule satisfies component 1 but fails component 2, it is categorically false.

2. Matrix Grid Solving Archetypes & Mathematical Formulations

Archetype A: Linear Row & Column Combinations

Pattern Name	Mathematical Formulation	Typical Manifestation
Row-wise Summation	$C_{i,3} = C_{i,1} + C_{i,2}$	The third column is the direct sum of the first two.
Row-wise Difference	$C_{i,3} = C_{i,1} - C_{i,2}$	The third column is the arithmetic difference.
Incremented Product Sum	$C_{i,3} = (C_{i,1} + 1)(C_{i,2} + 1) - 1$	$C_3 = C_1C_2 + C_1 + C_2$.
Linear Differential Balance	$R_1 - R_2 = R_4 - R_3$	Constant gap across paired alternating rows.
Linear Multi-Row Superposition	$R_4 = R_2 + R_3 - R_1$	Column accumulation with subtractive offset.

Archetype B: Multiplicative & Fractional Proportions

Pattern Name	Mathematical Formulation	Typical Manifestation
Direct Product	$C_3 = C_1 \times C_2$	Direct multiplicative factor.
Product over Divisor	$C_4 = \frac{C_1 \times C_2}{C_3}$	Cross-column quotient balancing.
Proportional Scaling	$R_k = \lambda \cdot R_{k-1}$	Row 2 is $0.5 \times R_1$; Row 3 is $3 \times R_2$.
Column Accumulation Scaling	$C_4 = \lambda \sum_{j=1}^{3} C_j$	Fourth column is a scalar multiple of column sum (e.g. $\lambda = 2$).
Fractional Mapping	$C_3 = C_1 \times \frac{p}{q}$	Fractional scalar producing an integer mapped to the alphabet.

Archetype C: Powers, Exponents & Polynomial Grids

Pattern Name
Exponential Degrees
Square Offset
Sum of Squares
Radical / Cube Root Sum
c-2.7,0,-7.17,-2.7,-13.5,-8c-5.8,-5.3,-9.5,-10,-9.5,-14
c0,-2,0.3,-3.3,1,-4c1.3,-2.7,23.83,-20.7,67.5,-54
c44.2,-33.3,65.8,-50.3,66.5,-51c1.3,-1.3,3,-2,5,-2c4.7,0,8.7,3.3,12,10
s173,378,173,378c0.7,0,35.3,-71,104,-213c68.7,-142,137.5,-285,206.5,-429
c69,-144,104.5,-217.7,106.5,-221
l0 -0
c5.3,-9.3,12,-14,20,-14
H400000v40H845.2724
s-225.272,467,-225.272,467s-235,486,-235,486c-2.7,4.7,-9,7,-19,7
c-6,0,-10,-1,-12,-3s-194,-422,-194,-422s-65,47,-65,47z
M834 80h400000v40h-400000z"/> + 3 C2
c-2.7,0,-7.17,-2.7,-13.5,-8c-5.8,-5.3,-9.5,-10,-9.5,-14
c0,-2,0.3,-3.3,1,-4c1.3,-2.7,23.83,-20.7,67.5,-54
c44.2,-33.3,65.8,-50.3,66.5,-51c1.3,-1.3,3,-2,5,-2c4.7,0,8.7,3.3,12,10
s173,378,173,378c0.7,0,35.3,-71,104,-213c68.7,-142,137.5,-285,206.5,-429
c69,-144,104.5,-217.7,106.5,-221
l0 -0
c5.3,-9.3,12,-14,20,-14
H400000v40H845.2724
s-225.272,467,-225.272,467s-235,486,-235,486c-2.7,4.7,-9,7,-19,7
c-6,0,-10,-1,-12,-3s-194,-422,-194,-422s-65,47,-65,47z
M834 80h400000v40h-400000z"/> = C3
Squared Differences

3. Geometric & Figure Puzzle Formulations

1. Triangles with Vertex Inputs

- **Additive Factor:** $\text{Center} = (\text{Top} + \text{Bottom-Right}) \times \text{Bottom-Left}$
- **Digital Sum of Product:** $\text{Center} = \text{DigitalRoot}(V_1 \times V_2 \times V_3)$
- **Pure Product Factor:** $\text{Center} = V_1 \times V_2 \times V_3$

2. Segmented Polygons & Ovals

- **Cross Determinant:** $\text{Center} = (\text{Left} \times \text{Right}) - (\text{Top} \times \text{Bottom})$
- **Propeller / Cross Product:** $\text{Center} = (TL \times BR) + (TR \times BL)$

- **Corner Degree Superposition:** $\text{Center} = TR^3 + TL^2$
- **Semiproduct with Constant Offset:** $\text{Center} = \frac{\text{Left} \times \text{Right}}{2} - k$

4. Alphabet-to-Number Positional Logic ($A = 1 \dots Z = 26$)

In Infosys puzzles, alphabets represent their forward English positional values:

$$\begin{array}{ccccccccc} A=1 & B=2 & C=3 & D=4 & E=5 & F=6 & G=7 & H=8 & I=9 \\ J=10 & K=11 & L=12 & M=13 & N=14 & O=15 & P=16 & Q=17 & R=18 \\ S=19 & T=20 & U=21 & V=22 & W=23 & X=24 & Y=25 & Z=26 & \end{array}$$

Common Alphabet Transformations

1. **Positional Difference Doubling:** $(\text{Top} - \text{Bottom}) \times 2 = \text{Center}$.
2. **Alphabet Arithmetic Progression:** Odd sequences ($1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23$).
3. **Paired Letter Multiplicative Sum:** $(L_1 \times L_2) + (L_3 \times L_4) = \text{Number}$.
4. **Alphabet Semi-Product:** $\frac{L_1 \times L_3}{2} = L_2$.
5. **Sum & Difference Vector:** $L_1 - L_2 = C_2$ and $L_1 + L_2 = C_3$.

5. Strategic 60-Second Solving Algorithm for Test Takers

When facing an unknown matrix or figure puzzle under test conditions, apply this prioritized sequence:

```
[Step 1: Check Magnitude Gradient]
├── Are numbers roughly similar?
│   └── Test Row/Column Sums, Differences, or Averages.
└── Are numbers exploding in size?
    └── Test Multiplication, Squares, Cubes, or Powers.

[Step 2: Check Dimensions & Symmetry]
├── 3x3 Grid: Check Row operations first, then Column operations, then Diagonals.
├── 4x3 Grid: Check if Row 4 is the sum or multiple of Rows 1, 2, 3.
└── 4x4 Grid: Check for parallel Arithmetic Progressions along rows and columns.

[Step 3: Test Digits & Non-Linear Functions]
├── Do numbers resist standard algebra?
│   ├── Test Digit Products: (Digit 1 * Digit 2) + ...
│   ├── Test Digital Root: Sum of digits until single digit.
│   └── Test Reverse Digits: |A - B| with reversed digit order.

[Step 4: Distractor Elimination]
├── Calculate parity (odd/even).
├── Check monotonic bounds.
└── Reject options that violate modular arithmetic.
```